

AADITYA SANIL

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EDUCATION

University of California, Berkeley

Berkeley, CA

B.A. Computer Science

Expected Graduation: May 2028

- GPA: **3.70** | Coursework: Data Structures, Principles of Data Science, Optimization Models, Discrete Mathematics, Probability, Computer Architecture.

EXPERIENCE

Imagine Games Network (IGN)

Aug. 2025 – Dec. 2025

Software Engineer Intern

Los Angeles, CA

- Engineered automated **Python** data pipelines, optimizing data quality by 15% and reducing execution cycles to **4 hours**.
- Developed high-performance feature selection microservices to process **20k+ sparse user intent signals**, improving system precision from 93% to 96%.
- Architected scalable backend infrastructure for high-volume data, supporting segmentation systems serving **millions of monthly active users**.

Martin Computational Speciation Lab

June 2025 – Present

Undergraduate ML Researcher

Berkeley, CA

- Reduced simulation latency by 83%** (60min to 10min) by refactoring the Julia backend into a **scalable K-buffer architecture** for high-throughput distributed execution.
- Architected system capacity for 100k+ concurrent entities**, implementing multi-variable biological parameters for complex population dynamics.
- Built Python data pipelines exporting 26 telemetry metrics per run for automated regression testing and analysis.

DreamWorks Animation

Feb. 2025 – June 2025

Technology Strategy Consultant (Contract)

Berkeley, CA

- Designed **system architecture roadmaps** for **Generative AI (Stable Diffusion)** integration across 3 departments to increase content production.
- Executed **performance profiling** on Diffusion models, optimizing for **low-latency inference** and compute resource allocation in real-time artist workflows.
- Guided cross-functional **scalability decisions** by presenting **quantitative throughput metrics** to senior leadership, securing a phased deployment strategy.

PROJECTS

BMRC Logistics System | *React, FastAPI, PostgreSQL, Docker* | [Live App](#)

Nov. 2025 – Present

- Engineered a scalable inventory system supporting **200+ concurrent users** using a **PostgreSQL** relational schema designed for strict data integrity.
- Developed **RESTful microservices** with FastAPI, implementing optimized **CRUD operations** for real-time tracking and dynamic search.
- Integrated **JWT-based authentication** and RBAC to secure backend endpoints, ensuring protected data access across the architecture.

AI Security: Provably Robust Classifiers | *Python, PyTorch, Optimization* | [GitHub](#)

Jan. 2026 – April 2026

- Simulated **adversarial attacks** to identify and mitigate **security vulnerabilities** in deep learning models.
- Architected a **formal verification framework** using optimization to provide **robustness certificates** for system reliability.
- Engineered a **security-aware training pipeline** to optimize the balance between **model accuracy** and system resilience.

ProteinLLMV1 — Protein Language Model | *nanoGPT, FastAPI, ESMFold* | [GitHub](#)

Dec. 2025 – Present

- Developed a custom 12-layer transformer model pre-trained via **Masked Language Modeling (MLM)** on UniProt sequences for protein structure analysis.
- Integrated **ESMFold** with a FastAPI backend to visualize 3D protein folding dynamics and confidence scoring.

TECHNICAL SKILLS

Languages: Python, Java, C, C++, Julia, R, SQL

Frameworks: FastAPI, Flask, React, Node.js

Data & ML: PyTorch, Scikit-learn, Pandas, NumPy, Matplotlib, Polars, DuckDB, CVXPY

Tools: Docker, Git, PostgreSQL, SQLite, Linux, Bash